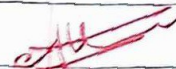
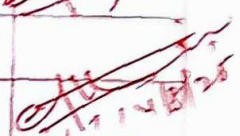
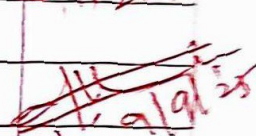
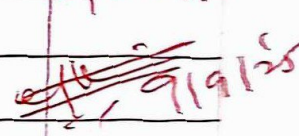
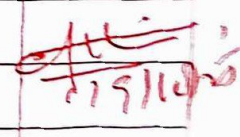
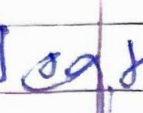
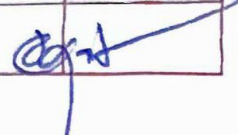
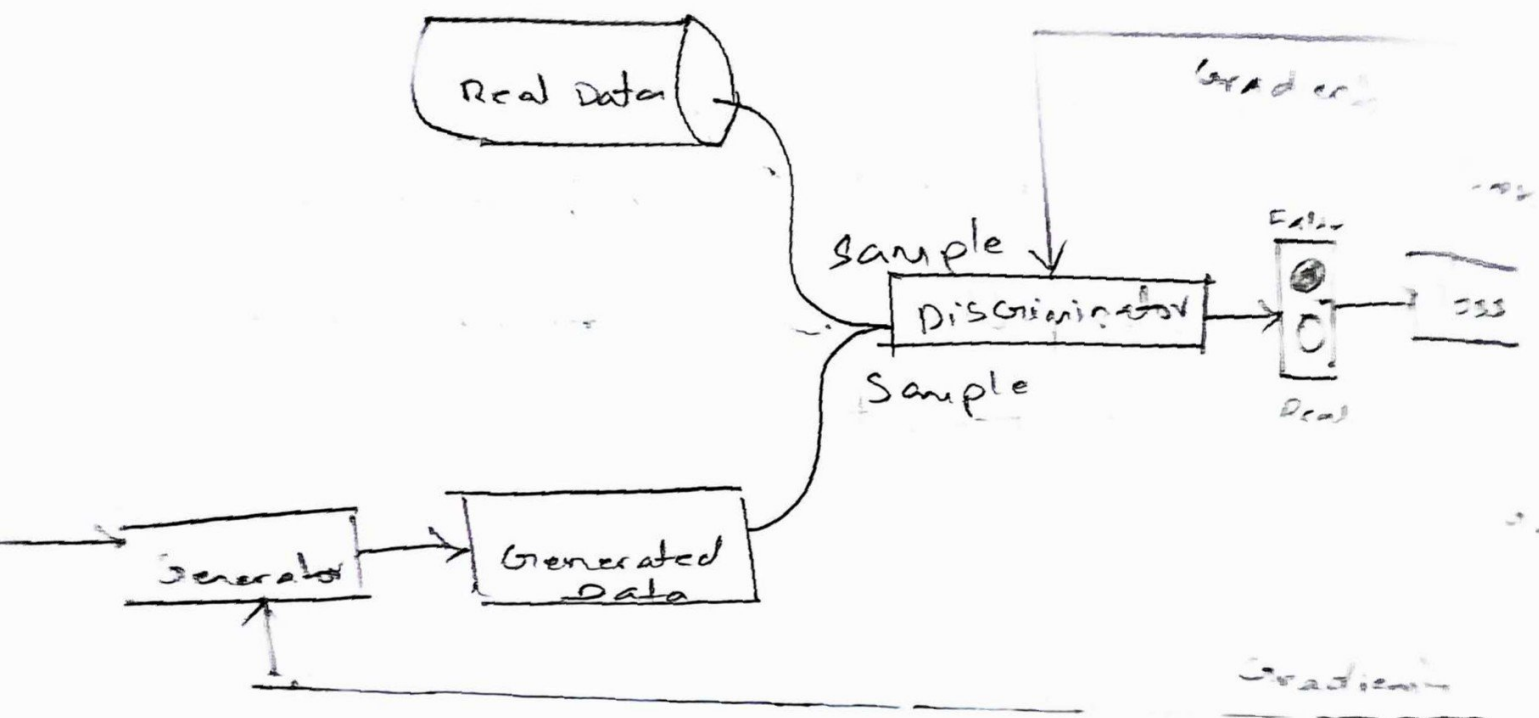


I N D E X

NAME: Kuberan STD.: A1 SEC.: A ROLL NO.: _____ SUB.: Deep learning

S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
1	24/7	Exploring Deep learning Platform		
2	31/07	Implement a classifier using open source dataset		
3	14/08	Study of the classifiers with respect to statistical parameters		
4	14/08	Build a simple feed forward neural network to register uc		
5	22/08	Study of Activation function and its role		
6	09/09	Gradient descent & backpropagation in deep neural networks		
7	16/09	Build a cnn model classify cat and dog image		
8	9/10	experiment of LSTM		
9	09/10	Build a Recurrent Neural Networks		
10	17/10	Perform Compression on mnist dataset using Autoencoder		
11	17/10	Experiments using variational Autoencoder (VAE)		
12	26/10	Implement a Deep convolutional GAN (DCGAN) to color image		
13	26/10	Understand the Architecture of a pre-trained Model		
14	26/10	Implement a pre-trained CNN model as a Feature Extractor		
15	26/10	Implement a yolo model for object Detection		

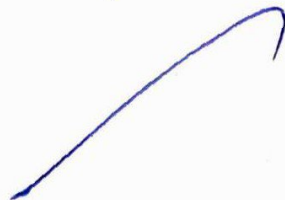
1) GAN Architecture



2) ~~Universal Random Noise~~

output

Epoch	Dloss	Gloss
1	1.372	0.873
2	1.102	1.2
3	0.987	1.4
4	0.84	1.6
5	0.73	1.9
6	0.70	2.1
7	0.66	2.3
8	0.64	2.4
9	0.61	2.5
10	0.59	2.7



3. Define Discriminator network
 - Input: real or generated image
 - Convolution layers + Leaky Relu
 - Output: probability (real or fake)
4. Compile Discriminator (binary cross entropy, Adam)
5. Create GAN model by connecting Generator $\rightarrow$ Discriminator
6. End

Observation:

1. Generator began producing blurry patterns initially
2. After several epochs, clearer shapes and textures began forming
3. Discriminator accuracy fluctuated due to adversarial training
4. The Balance between generator and discriminator was crucial for stable training

Result:

Q. A DCGAN mode was successfully implemented and trained.